

7 DEWATERING AND GROUNDWATER CONTROL

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7.1 IMPACT OF GROUNDWATER ON CONSTRUCTION

Whenever excavation must take place below the water table, groundwater affects the project. It affects the function and design of the facility, and the cost of its construction. Groundwater is a frequent cause of disputes between the owner and the contractor. Dewatering by unsuitable methods can under some conditions cause damage to adjacent properties, and result in third party litigation. Under some conditions dewatering may be harmful to the environment. Activities involving groundwater are closely regulated in many areas. The process of obtaining permits is often tedious, and sometimes authorities require special procedures that can be expensive.

Engineers responsible for all the phases of a project, from initial planning and budgeting through final construction need to be aware of the potential impact of groundwater so that their decisions will be effective. This chapter describes problems that groundwater has created, and presents methods that have proven effective in alleviating those problems. References are provided to more detailed treatment of the subject.

7.2 DESIGN OF STRUCTURES BELOW THE WATER TABLE

Factors that planners and designers should consider include the following.

Waterproofing More elaborate methods are necessary, particularly with very deep structures, to insure watertight integrity. Flexible membranes are recommended so that minor structural cracking does not cause leaks. There are many cases where expensive underground space has been rendered unusable by seepage. Postconstruction corrective measures such as grouting or permanent dewatering may be costly.

Hydrostatic Pressure Slabs and walls must be designed to carry both soil loads and water pressure. Where column spacing is wide, slab strengthening can be a major cost.

Uplift The weight of the structure must be sufficient to resist buoyancy. If the calculations credit the weight of the upper floors of a building for example, or the equipment in a lift station, then temporary dewatering must be continued until the necessary weight is in place. There have been instances where structures have heaved owing to premature release of groundwater control.

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Relieved Structures Drydocks, sewage treatment tanks, and lift stations are often provided with drainage systems to lower the hydrostatic pressure when

they are emptied. Building basements have been designed with dewatering systems that permanently depress the water table. A relieved design can sharply reduce first cost, but the drainage systems demand special methods and annual costs must be considered (Section 7.10).

Depth versus Cost Often a deep excavation is an option. The planner may be considering underground parking in an urban area, or the designer may be comparing the cost of piling to a deep floating foundation. For such decisions a reliable estimate of dewatering cost is essential. There have been cases where a major design change became necessary after construction began, because of unexpectedly difficult dewatering.

Permanent Effects on the Groundwater Regime Structures below the water table can affect the groundwater regime, sometimes undesirably. A cut-and-cover subway may create a dam in the path of natural groundwater flow, raising water tables upgradient and depressing them downgradient. Relieved structures can permanently depress the water table. Sewers placed in gravel bedding can modify groundwater flow patterns. If these factors cause undesirable side-effects (Section 7.9) corrective measures can be incorporated during construction. Drainage may be advisable around or beneath cut-and-cover subways. Relieved structures can be surrounded with cutoffs. Gravel bedding around pipes can be plugged at intervals with clay or concrete. Such measures are more expensive after construction.

7.3. METHODS OF GROUNDWATER CONTROL

The many methods that have been developed for the control of groundwater fall into three basic categories.

Open Pumping is the process in which as the excavation is advanced the water is allowed to flow in and is collected in ditches and sumps, then pumped away. The open pumping method is lowest in direct dewatering cost, and is viable under certain conditions. Under other conditions it can seriously hamper other operations, and can result in catastrophic failure.

Conditions favorable for open pumping include shallow subgrades, no more than a few feet below water table, and stable soils of low to moderate permeability such as dense, well-graded tills, clayey sands, firm clays, or fissured rock. There must be

room for relatively flat slopes, or the excavation must be supported.

Open pumping is not effective in nonplastic sands and silts, since boiling can loosen the soil at subgrade, harming its bearing capacity. When excavating into an aquifer of high transmissibility, a large initial rate of pumping will occur during storage depletion. Open pumping will be cumbersome, unless the excavation is carried out very slowly. Where there are existing structures nearby, loss of soils due to open pumping may cause damage. If artesian pressure exists in a confined aquifer below subgrade, the condition cannot be relieved by

open pumping. Conditions that favor or preclude open pumping are more fully discussed in Powers (1981).

During project planning, open pumping should not be budgeted for unless there is reasonable probability that conditions favorable to the method exist. When applying the method on a project underway, continuous observation is recommended to determine whether an undesirable situation is developing. A contingency plan incorporating one of the more positive methods of groundwater control is advisable.

Predrained water table

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Predrainage is the process of lowering the water table prior to excavation, using one of several methods available.

Deep Wells with individual pumps are most suitable for free-draining soils, and where the bottom of the aquifer is well below the subgrade of the excavation (Fig. 7.1). In this situation the yield per well in the dewatered condition is high, and relatively few wells are required. Since the cost per well is high, if the wells must be spaced very closely to accomplish the desired result the method may not be cost-effective.

Wellpoint Systems utilize groups of closely spaced wellpoints connected to a common suction header and pump. The cost per wellpoint is low, and in stratified soils where the spacing must be close the installation cost is lower. However, wellpoints depend on atmospheric pressure to lift the water to the pump, and the amount the water can be lowered with the method is limited to about 15 to 18 ft (5 to 6 m) at sea level, less at higher elevations. For greater drawdowns, multistages of wellpoints are required.

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Fig. 7.1 Predrainage with deep wells.

Predrained water tables <=

Normal water tables

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Fig. 7.2 Combination of deep wells and open pumping. Most of the water volume Q is controlled by wells. Gravel or other slope treatment

may be necessary where residual seepage escapes.

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Normal water table

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Predrained water level

Well Se

Wellpoints

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Fig. 7.3 Combination of deep wells, wellpoints, and open pumping.

Ejector Wellpoints incorporate a nozzle and venturi in the riser pipe. This method of pumping, similar to the domestic jet pump, eliminates the suction lift limitation of conventional wellpoints. Water tables have been lowered 100 ft or more using a single stage with this method. The pumping efficiency is poor, and for volumes in excess of 1000 gpm ejectors are rarely cost-effective.

Cutoff and Exclusion encompass a variety of methods that have been developed to obstruct entry of water to the excavated area.

Vertical cutoffs include driven steel sheet piling (NAVFAC DM-7, 1971, and Chapter 12) and concrete diaphragm walls (Xanthakos, 1979) built in panels in a bentonite slurry. The slurry trench (EPA, 1979, and Chapter 20) normally involves a continuous trench under bentonite slurry, backfilled with a mixture of soil and bentonite, or with bentonite/cement grout.

Permeation Grouting has been used to provide partially effective cutoffs and the newer jet grouting process has produced effective cutoffs in some soils (Baker, 1982; Karol, 1983). Ground freezing (Maishman, 1975; Sanger, 1968) has been successful as water cutoff and ground support under a wide variety of conditions.

Horizontal water cutoffs have been constructed by the tremie seal method (NAVFAC DM.-7, 1971) and by ground freezing.

Water can be excluded from tunnels and shafts during excavation by compressed air, the slurry shield, and the earth- pressure shield (Bickel and Keusel, 1982; Pequignot, 1963; Richardson and Mayo, 1975).

Each of the methods of cutoff and exclusion has characteristics that make it more or less suitable under various ground conditions and project requirements (Powers, 1981).

Methods in Combination

Given the wide variety of methods for groundwater control and the many variations in project requirements, it is not surprising that frequently the most

cost-effective solution is one that combines two or more of the methods on a single project.

Deep Wells and Open Pumping Figure 7.2 shows an excavation in stratified soil where open pumping alone is not a viable method. To fully predrain the water would require a great many wells. However, a well array on reasonable spacing, say 20 to 40 ft, will drain the more permeable strata to the extent that residual seepage can be controlled safely with sumps. This combination is most effective where recharge is remote and

storage can be depleted by lengthy pumping in advance of excavation.

Deep Wells, Wellpoints and Open Pumping Figure 7.3 shows a deep excavation that penetrates a bed of impermeable clay. A large number of wells would be required to predrain the water close enough to the clay that open pumping would be viable. A lesser number of wells have been installed, to lower the water table to within suction lift of a single-stage wellpoint system. The residual seepage past the wellpoints is handled with open pumping.

Steel Sheet Piling and Wellpoints Figure 7.4 shows an intake structure in open water. It may be difficult and costly to drive

PLAN Wellpoints Water Steel sheet piling surface nn Vv River bed Subgrade Sand and gravel f i Bedrock SECTION Fig. 7.4 Intake structures, combination of cutoff and predrainage.

A DW #1 DW #2 A

2 i ES 4 4

PLAN b

Normal artesian head, lower aquifer _ iv ~< Well >

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Steel sheet piling

Normal water table (upper ~]

Subgrade uquiter) wo

ZClay

Wu

SECTION A—A Fig. 7.5 Cutoff and pressure relief.

the sheeting to rock to provide cutoff. A wellpoint system is used to control the seepage under the toe of the sheets, providing a dry and stable subgrade, and assuring passive strength of the sand to support the toe of the sheeting. In some such cases, the most cost-effective combination may be to extend the toe of the sheeting deeper and use wells rather than wellpoints.

Cutoff and Pressure Relief Figure 7.5 shows steel sheeting extending to a clay layer. Artesian pressure beneath the clay presents a risk of heave. Deep wells are used to relieve the pressure.

7.4 FIELD PUMPING TESTS

Every geotechnical investigation should include the basic observations of groundwater conditions described in Section 1.10. If a potential groundwater problem has been identified, more elaborate procedures are advisable, such as additional borings completed as observation wells, sieve analysis of samples, and research on dewatering and water supply history in the area. When it appears that groundwater may significantly affect the design of the structure or the cost of the project, or result in disputes or third-party claims, then a field pumping test should be considered.

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Fig. 7.6 Typical pumping test array. If recharge or barrier boundaries are suspected, lines of observation wells may be arrayed in several directions.

A typical array is shown in Figure 7.6. The test well is designed to stress the aquifer sufficiently to produce meaningful drawdowns in the observation wells (Powers, 1981). The analysis techniques assume that the aquifer is homogeneous and isotropic within the zone of pumping influence, that all water pumped is from aquifer storage, and that it is instantaneously released. Natural aquifers almost never meet the requirements, so the analyst's major challenge is to interpret from anomalies in the data the characteristics of this aquifer and how it departs from the ideal. For this purpose the semilog plots of the Jacob modification are most suitable, and will be discussed here. More elaborate methods are available, and are useful in special conditions. Certain criteria on length of the test must be fulfilled before the Jacob modification is valid (Driscoll, 1986; Powers, 1981).

The test must be operated long enough to identify recharge or barrier boundaries, or delayed storage release. In a water table aquifer, 7 days is recommended. In an artesian aquifer, 24 hours is often sufficient. The pumping should be continuous at a constant rate.

Water levels are recorded periodically prior to pumping to observe any natural fluctuations (tides, river stages, etc.) When pumping begins, early water levels are recorded on a frequent schedule, typically every few minutes for an hour. Readings can then be spaced out logarithmically with time. The drawdowns

are plotted against the log of time for each observation well, and versus log of distance for all observation wells at various times. Figure 7.7 shows typical plots in an ideal aquifer. It is good practice to make the plots as the test proceeds, so that it can be decided whether enough information has been gathered before terminating the test. Recovery readings are taken when pumping stops, on a frequency similar to pumpdown.

At some point into the test, samples should be recovered for chemical and bacteriological analysis.

The plots of Figure 7.7 can be used to calculate the key parameters of the aquifer, transmissibility T and storage coefficient C . Transmissibility is defined as the ability of the aquifer to transmit water, per unit width normal to flow direction, under unit hydraulic gradient. In US units, T is measured in gallons per day per foot (gpd/ft). The storage coefficient is the yield of the aquifer in unit volume of water per unit volume of soil dewatered or pressure relieved. It is unitless. Referring to Figure 7.7:

Q = constant flow rate, gpm

h = observed drawdown from static, feet Δh = drawdown difference per log cycle, feet

t = time since pumping started, minutes

t' = time since pumping stopped, minutes

h_r = residual drawdown during recovery period

t_0 = zero-drawdown intercept on log-time plot

r = radius from pumped well to observation well, feet R_y = radius of influence, zero-drawdown intercept on log

distance plot

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From the drawdown versus log time plot (Fig. 7.7a),

$T = \frac{Q}{4\pi h} \left(\frac{2.303}{\Delta h} \right) \quad (7.1)$
 From the drawdown versus log distance plot (Fig. 7.7b),
 $T = \frac{Q}{4\pi h} \left(\frac{2.303}{\Delta h} \right) \quad (7.3)$

From the recovery plot of residual drawdown versus log t/t' (Fig. 7.7c),

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$T = \frac{Q}{4\pi h} \left(\frac{2.303}{\Delta h} \right) \quad (7.5)$

AS (7.5)

In a uniform aquifer, transmissibility can be defined as the

product of the permeability k in gpd/ft and the aquifer thickness B in feet:

(7.6)

$$T=kB$$

Conversely, the permeability can be calculated from the transmissibility determined as above from a pump test, by dividing by the aquifer thickness as determined from boring logs. But this is only for a uniform aquifer. Most aquifers have seams with quite variable k , and calculated k is some sort of average value. It can distort the dewatering analysis (Section 7.5), and must be used with judgment.

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Fig. 7.7a Drawdown versus log time t.

0 aL R, = 1900 ft Ow-3 4h ow-2 im Q = 150 gpm a ow. | a" (= 720 min (12 hr)
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24750 gpo/ft — 24750 x 720 pe * 4790 x (1900) = 1x10⁷ boy piiiitil bop iiii
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t Fig. 7.7c Residual drawdown versus log a

An advantage of the Jacob plots is that any departure from the ideal straight line is readily apparent, and the analyst is forewarned. He then must compare the values of T and C , obtained from the various plots and attempt to judge from the shape and offset of the curves the nature of the anomalies in the natural aquifer with which he is dealing. Note that in an ideal aquifer that fulfills all the assumptions, the values of T and C , as calculated from all three plots of Figure 7.7 are the same. When they are not, there is a problem. The shape and offset of the curves provide clues. Figure 7.8 illustrates some distortions commonly encountered.

Figure 7.8a shows the effect on time plots of recharge and barrier boundaries. Figure 7.8b shows the effect of delayed storage release. If T is calculated from the portion of the curve that has been flattened by delayed release, the value can be grossly in error. Figure 7.8b also illustrates the effect of a pumping interruption. Note the difficulty if no observations of drawdown were being made during the interruption. Figure 7.8c illustrates the effect of tides.

Figure 7.8d shows the effect on distance plots of recharge and barrier boundaries, and Figure 7.8e shows the effect of partial penetration. The partial penetration effect becomes insignificant at a radius r roughly equal to 1.5 times the saturated thickness of the aquifer, B . A quick scan of the variations in shape of these distorted curves and in their

(a) Aquifer with recharge - ——— 2 me (Aquifer with = barrier boundary a doi
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Radius r, ft

(e)

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Partial penetration

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Radius r, ft

Fig. 7.8 (a), (b), (c) Time plot distortions. (d), (e) Distance plot distortions.

zero-drawdown intercepts will demonstrate how far calculated values can be in error.

7.5 DESIGN OF PREDRAINAGE SYSTEMS

Well Systems For the design and cost projection of a typical well system, it is necessary to estimate the total Q to achieve the required drawdown, and the yield Q_w , of each well in the dewatered condition. These determine the number N of wells

required. Large total Q does not necessarily mean high cost. A free-draining aquifer may be dewatered at reasonable cost with a few large wells, whereas a project in stratified soils may require a great many wells, perhaps in combination with wellpoints, and although the total Q is less the cost may be greater.

In addition to the number and capacity of wells, it is necessary to estimate the area within which significant drawdown will occur, since this determines whether side-effects will have an impact on adjacent properties. It is also advisable to make

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a judgment on what conditions will exist within the excavation, based on soil descriptions and laboratory testing. If an impermeable stratum exists above subgrade, will perched water seep in over it, eroding the slopes? (Fig. 7.2). If the impermeable material is clay, the problem may be manageable; but if it is nonplastic silt, stability problems may be severe. Such considerations may affect the choice among water-control methods discussed in Section 7.3.

Mathematical Models

Confined Aquifer In simple aquifer situations, a suitable design approach is to model the excavation as a single well, with effective radius equivalent to the well system, to estimate the total Q . The problem illustrated in Figure 7.5 will serve as an example of the design procedures.

A rectangular structure of dimensions a and b is to penetrate through an upper aquifer of sand to a foundation in clay. Steel sheeting will be used to cut off the water in the upper aquifer, and to support the sides of the excavation. However, the clay is thin, and another aquifer beneath it must be pressure-relieved to protect the excavation from bottom heave.

It is proposed to surround the cofferdam with a system of pumped wells to relieve the lower aquifer. If we consider the well system as a single well of equivalent radius r_w , then

no (7.7) r_w

$$Q = \frac{2\pi k B (H - h_w)}{\ln(R_o/r_w)} \sim 2.29 \ln R_o/r_w$$

and

Q (7.8)

where

Q = total yield of the system, gpm

k = permeability, gpd/ft²

B = thickness of aquifer, feet R_o = radius of influence, feet

H = initial head in the aquifer, feet

w = final head at the equivalent well

Selection of values to use in Equation 7.8 requires judgment, and is based on information from many sources, including a field pumping test, the boring logs, data on surface hydrology (to evaluate recharge from infiltration, and from surface water bodies), and from data on groundwater hydrology (to evaluate recharge from other aquifers). Inexperienced judgment in the selection of values for Equation 7.8 has resulted in gross error in analyses of this type.

The product kB is of course the transmissibility T as determined from the field pumping test. B can be estimated from boring logs, and an average k can be calculated from Equation 7.6. Experience shows that values of T and k are frequently misinterpreted (Powers, 1986). In most pumping tests, the indicated values are for horizontal permeability k_h . This value is only appropriate when flow to the dewatering system will be essentially horizontal. In sedimentary soils, the ratio of horizontal to vertical permeability, k_h/k_v , is rarely less than 3, and can be 10, 100, or larger. If flow to the dewatering system is other than horizontal, use of k , in Equation 7.8 introduces error that can be significant. For example,

if the upper water table aquifer in Figure 7.5 were to be dewatered, the flow would be both horizontal and vertical. A procedure that has given reasonable estimates in such a situation is as follows: knowing k , from a pumping test, then k , is estimated from a study of the boring logs, and the general geology.

The value used in the mathematical mode is the isotropic permeability k_i .

$$k_i = \frac{k}{k_y} \quad (7.9)$$

The radius of influence R_y can be defined as the horizontal limit of the zone that will be affected by pumping. It is the zero-drawdown intercept on a Jacob distance plot (Fig. 7.7b). If all water pumped was from aquifer storage, then R_y could be estimated by rearranging Equation 7.4:

$$R_y = \sqrt{\frac{2.25 T t}{S}} \quad (7.10)$$

Thus, R_y without recharge or barrier boundaries is a function of the transmissibility and storage coefficient, and expands with the square root of time. In many dewatering situations, R_y expands to a value less than indicated by Equation 7.10, because of recharge of surface water or leakage from other aquifers. Judgment must be made on any potential sources of recharge so that an equilibrium R_y can be estimated. Since R_y is a log function, a precise estimate is not necessary.

Within the operating period of a pumping project R_y can vary. During heavy rains it can contract. Along volatile rivers when spring floods cause broad inundation of the flood plain, R_y can shrink dramatically, and Q increases. Unless provision has been made in the design of the dewatering system, it may be overpowered. The U.S. Army Corps of Engineers monitors the stages of many rivers. Hydrographs can be researched, and topographic maps studied to evaluate the possibility of major inundation. In many areas, flood insurance maps are available to show the inundation during storms of various frequencies.

Initial head H in the aquifer can vary seasonally with precipitation or the stage of an adjacent river, and can be affected by other pumping in the vicinity. Appropriate safety factors should be included.

Final head h_w at the radius of the equivalent well must be lower than final head H that is desired under the excavation. The differential $h - h_w$ is a function of k , B , Q , and the geometry of the system.

When appropriate values have been estimated for the variables, total Q can be estimated from Equation 7.8. An estimate is then made of Q_w , the yield of an individual well in the dewatered condition, and the required number of wells N can be calculated:

$$N = \frac{Q}{Q_w}$$

Well yield Q_w is best evaluated from pump test data. In extrapolating the test results, adjustments must be made for conditions at the well that are different from those during the test. A factor of major significance is s , the saturated

thickness of aquifer in contact with the wall of the borehole. In uniform aquifers, Q_w is directly proportional to b_w .

The phenomenon of well loss has been studied by Walton (1970) and others. No well can provide frictionless entry, so some well loss always occurs. Poorly constructed wells can have very high well loss. Unsatisfactory practices include the use of bentonite as a drilling fluid, inadequate flushing of the borehole to remove drilling detritus before completion, low-quality wellscreen, filter packs that are inappropriate to the aquifer or that have been ineffectively placed, and insufficient well development (Powers, 1981; Driscoll, 1986). Well loss during a pump test can be evaluated by estimating the drawdown at the radius of the borehole from the Jacob drawdown—distance plot, and comparing it with the pumping level in the well. It can also be estimated from step drawdown tests conducted before or after the constant rate test (Walton, 1970).

N (7.11)

In the absence of a pumping test, Q_w can be estimated crudely with a modified version of the Sichart formula (Powers, 1981),

$$Q_w = (0.04)I_r/k \quad (7.12)$$

where

r_w = the borehole radius in inches

b_w = the saturated aquifer thickness in contact with the borehole

k = estimated permeability in gpd /ft*

Sichart estimates are sometimes reasonably close, but can be far from actual, in either direction.

Water Table Aquifer Analysis of dewatering in a water table aquifer is quite different from pressure relief in a confined aquifer. The water table aquifer shown in Figure 7.1 has a variable phreatic surface, and its saturated thickness H changes during pumping. Q can be calculated from the relationship

$$k(H_0 - h_2) = \frac{Q}{2\pi} \ln \left(\frac{R_0}{r_w} \right) \quad (7.13)$$

Values for the variables are estimated as in the discussion above. However, in water table aquifers an average k calculated from Equation 7.6 can introduce significant error. The aquifer will be partially dewatered. If the remaining saturated thickness h has a higher k than the average, higher Q must be expected. It is of note that when the deeper k is lower than average, the dewatering is more difficult. Q_w is lower, more wells are required, and although the total Q is less the cost may be significantly higher.

Other mathematical models that have proven useful in analyzing various aquifer situations are given in Figure 7.9.

The method of cumulative drawdowns can provide more accurate analysis of the confined aquifer in Figure 7.5 than the simple mathematical model. Accu-

racy is not the principal advantage however, since with the variables imprecisely known, precision in analysis is of limited value. The chief advantage of cumulative drawdowns is the method's ability to analyze many points within the flow regime.

It is assumed in Figure 7.5 that

$a = 60$ ft

$b = 40$ ft $H - h = 27$ ft under general subgrade $H - h = 30$ ft under the pit

From the pump test of Figures 7.6 and 7.7, values are estimated as

$kB = 25\,000$ gpd/ft

$R_y = 2500$ ft at pumping time t when pressure relief must be accomplished.

$Q_{,,} = 175$ gpm in the desired condition

A plot is constructed showing the effect of each well (Fig. 7.10). The slope of the curve can be calculated by rearranging Equation 7.3:

mine

7 (7.14)

The drawdown at any point of interest is the summation of the drawdowns due to each well at its radius from the point, as read from Figure 7.10. The cumulative drawdowns are shown in Table 7.1. Note that the well array is asymmetric, to provide greater drawdown at point y under the pit. The method is useful in checking the predicted value of $Q_{,,}$. Aquifer drawdown at

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Radial flow in artesian aquifer

$kB(H - h,)$

$2 = 559 \ln(R_o/r,)$

Radial flow in water table aquifer

$\frac{1}{2} k (H^2 - h^2) 458 \ln (R_o/r,)$

Linear flow in

in

$| \times 1440L$

i_j

Linear flow in

— water table aquifer

ii $H Q \times$

$$i k(H^* - h^*)$$

$$\{ , ^{TM}$$

“

Fig. 7.9 Useful mathematical models: s = distance to line source (feet); x = unit length of drainage trench.

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it pid fy pit bo iil 1 pin) 1.0 10 100 1000 10000

Radius r , ft Fig. 7.10 Cumulative drawdown analysis.

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TABLE 7.1 CUMULATIVE DRAWDOWN ANALYSIS.

Point x Point y Point z DW #1

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Well No.

DW-1 25 74 10 88 20 770610 «12.5 DW-2 25 74 10 88 35 69 15 8.2 DW-3 276
72 41 66 58 60 651 6.3 DW-4 27.6 72 «41 66 43 65 46 6.4

Total 5 29.2 30.8 27.1 33.4

well #1 has been calculated in Table 7.1, using an effective radius of the well as 1 ft. If the remaining head h_w in the aquifer is less than the aquifer thickness B , the well may not yield what was estimated.

The cumulative drawdown method is suitable for confined aquifers, and gives reasonably reliable results in water table aquifers where the initial saturated thickness is not reduced by pumping by more than 20 percent.

Flow Net Analysis is useful when the geometry of the problem is too complex to apply a mathematical model or the cumulative drawdown method. Cedergren (1968) discusses the construction and application of flow nets. Complex situations in either plan or section can be analyzed. Difficulty occurs when the problem is complex in three dimensions. Perhaps some future geohydrologist whose hobby is three-dimensional chess will develop a tool for such problems.

Fragment Analysis developed by Harr (1977) is a mathematical approach similar to flow nets. Harr derived a group of relationships that estimate conditions in various fragments of a flow regime. He made several simplifying assumptions, while pointing out that the error thus introduced is well below the errors caused by variation in permeability and other factors. Fragment analysis is most useful when multiple iterations are required to find an optimum solution, and repeated

drawing of flow nets is cumbersome. For example, if in Figure 7.3 it is desired to examine the cost-effectiveness of various penetrations of the steel sheet piling, fragment analysis can be useful. The method shares with flow nets the difficulty presented by a problem in three dimensions.

Computer Analysis has proven useful when aquifer situations are complex. Modeling techniques have been developed by a number of investigators (Akin, 1982; Driscoll, 1986). They can be particularly effective in saving time when multiple iterations are required. The game of “what if” is helpful in estimating the potential impact of conditions other than those assumed. What if another nearby aquifer is recharging the one under study? What if k is different from that assumed? What if the adjacent river experiences a storm of hundred-year frequency during the dewatering, and inundates the flood plain? Computer models with their remarkable speed can save considerable time in evaluating such possibilities. In practice, without the computer the analyses frequently just do not get accomplished. Experience demonstrates, however, that there is an occupational hazard with computer analysis. One becomes so fascinated with the method that not enough time and effort is devoted to making the judgments on aquifer parameters discussed above. If those judgments are poor, the result may be in serious error, no matter how sophisticated the analysis method.

7.6 MANAGING GROUNDWATER CONTROL

Where a significant groundwater problem exists, its successful solution depends on each participant in the project carrying out his functions in a responsible manner. This section offers suggestions on effective performance of the various functions, based on observations of past projects where groundwater created unnecessary difficulty, and observations of those projects where skilled management kept the groundwater problem within control.

The owner has perceived the need for a facility and has arranged its financing. Working through his consultants he seeks a contractor who will build a structure of the desired quality on schedule and within budget, and without disputes.

The geotechnical engineer must be alert for groundwater indications during the initial investigation (Section 1.10). Where a problem with significant potential appears, he should recommend a budget for closer study (Section 7.4).

The planner faces chronic budget restraints in the initial stage. But if his decisions are to be sound, he must have a reliable evaluation of the cost and impact of the water problem. Some decisions can be tentative until more data is available. Underground parking may or not be provided; piling or a deep mat foundation can be options; a lift station and force main may be substituted for a deep gravity interceptor. But when planning an underground metro, the options are limited. The planner's decision when to expand the groundwater investigation is based on many factors, and must frequently be a compromise.

When the role of the designer becomes dominant, tentative decisions must be made firm. The groundwater study must be essentially complete so that choices

between options can be made, based on reliable estimates of cost and other factors. A key decision is how to allocate responsibility for controlling the water. If such decisions must subsequently be reversed because they were based on inadequate information, costs escalate.

Permits Groundwater activity in many jurisdictions is regulated by the authorities. Permits may be required for drilling wells, for extracting groundwater and for releasing it to the environment. Sometimes hearings are required, which can be time-consuming. The permit requirements should be investigated well in advance of advertising for bids, to avoid unexpected delay. If the contractor is required to obtain some of the permits, it is good practice to alert bidders of the fact, and discuss the proposed plan with the regulating authorities ahead of time.

Contractor-designed Systems for groundwater control are normal practice, for good reasons. Water control affects other operations, such as excavation, ground support, and scheduling. Given latitude, the bidder can use his ingenuity to be more competitive. The contractor's operations must be regulated, however, to insure that the foundation properties of the subsoil will not be impaired, that the structure can be built safely and on schedule, and that adjacent structures will not be endangered. A recommended technical specification is:

Control of groundwater shall be accomplished in a manner that will preserve the strength of the foundation soils, will not cause instability of the excavated slopes, and will not result in damage to existing structures. Where necessary to this purpose, the water will be lowered in advance of excavation by wells, wellpoints, or similar methods. Open pumping will not be permitted if it results in boils, loss of fines, softening of the subgrade, or slope instability. Wells and wellpoints will be installed with suitable screen and filters so that pumping of fines does not occur. Discharge will be arranged to facilitate sampling by the engineer.

When a potential problem has been identified it is good practice to draw bidders' attention to it; for example, if the soil at subgrade is sensitive to upward seepage, if there is deep artesian pressure, or if the aquifer is sensitive to pumping interruptions.

The contractor should be required to provide a submittal of his plan for groundwater control, prior to beginning its installation. The engineer reviews this to assure it is in accordance with the specifications and with good practice. The review and approval does not release the contractor from his responsibility for the plan's adequacy. The contractor when preparing his submittal usually has available only the prebid information. On complex projects, the specifications sometimes require a second submittal. After the system has been installed, the contractor must demonstrate by test operation that the system will accomplish its purpose.

Owner-designed Dewatering Systems are sometimes effective, but more often they create problems. The designer has only the prebid information at hand, and tends to be overly conservative. Where the contractor is responsible he can

develop more data as the installation proceeds. Each well becomes a test boring, each pumping test adds insight. The contractor has incentive to locate his wells skillfully and build them of good quality to minimize his costs. If he is paid for each well the incentive is lost. With owner-designed systems skilled inspectors are necessary, and such specialists may not be readily available. Perhaps the most pernicious problem with owner design is mixed responsibility. If the control is a combination of predrainage and open pumping (Fig. 7.2) disputes arise over whether sufficient wells were provided to make the open pumping manageable.

The minimum system concept has been applied in some cases with satisfactory results. The specifications require, for example, that a minimum number of wells be installed, but the contractor is made responsible for the adequacy of the system, including additional wells that may be necessary. The owner is assured against overoptimism by the contractor, which could result in schedule delays or other problems. The contractor has incentive to build high-quality wells, to avoid extra expense. Any open pumping is clearly his responsibility.

The contractor's options in executing the dewatering are several. He may buy or rent the dewatering equipment and install it with his own forces, a frequent practice on routine projects. Where the dewatering is complex he can engage a specialty subcontractor to execute the work. He may invite proposals on a turnkey subcontract, where the specialist undertakes to install and remove a guaranteed system for a lump sum, and operate and maintain it for a price per unit time. Or the contractor may elect to subcontract part of the work, such as well drilling, on a unit price basis, and retain the risk.

7.7 DISPUTES OVER GROUNDWATER CONTROL

Groundwater is involved in a high percentage of the disputes that occur in underground construction, whether between the contracting parties or with third parties. Disputes can escalate the cost of a project, and delay its completion. Some number of disputes is inevitable, given the risks of the underground, but with preparation many disputes can be equitably settled in timely fashion, avoiding costly litigation. This section suggests procedures to reduce the probability of disputes, and to expedite the settlement of those that occur. It is based on observations of disputes that have occurred.

The primary preventive is an appropriate geotechnical investigation (Sections 1.10 and 7.4). When a problem has been

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identified and evaluated it is less likely to result in a dispute. The data should be provided accurately and completely to potential bidders.

The differing site condition clause used on federally funded projects and many others accomplishes two purposes. It avoids contingencies in the bids, since the contractor is assured that in the event of unexpected conditions he will receive an equitable adjustment. It also defines clearly what an unexpected condition is. The Federal clause (DOT, 1976; ASCE, 1988) has a large body of

legal precedent that enables the parties to understand what has been agreed to. To qualify for an adjustment under this clause the condition must be different from that portrayed in the contract documents, and also different from what an experienced contractor would expect. The procedure for reporting the condition is detailed.

Substantive progress has been made in developing procedures to avoid disputes among the contracting parties (ASCE, 1988). These include the issuance of a Geotechnical Design Summary Report, which provides the bidders with not only the data from the investigation but the opinions and conclusions of the owner's consultants, on which the design was based. A Dispute Review Board is recommended to facilitate settlement of disagreements as early as possible.

If the geotechnical investigation identifies conditions where the groundwater control may affect third parties, preventive measures should be considered (Section 7.9). It has been common practice for owners to place the responsibility for the side-effects of dewatering on their contractors. If the side-effects are beyond the contractor's control, the practice is not recommended. Some courts have held for example, that if in the planning of the project the owner and his engineers contemplated lowering the water table, and the process subsequently causes injury to third parties, the owner is at least partly responsible.

7.8 COST OF GROUNDWATER CONTROL

Groundwater does not lend itself to rule-of-thumb estimating. Costs per gallon of water pumped, per cubic yard of excavation below the water table, or per lineal foot of trench dewatered, can all vary by an order of magnitude or more. Even the cost per well varies surprisingly. On a rapidly advancing trench excavation where the well is in operation for less than a week and the pump, screen, and casing are repeatedly reused, the cost per well installed is quite low. A well for long-term operation in an urban street drilled with difficulty and requiring sophisticated surface accessories can cost 50 times as much. Given these variations it is advisable to cost each project on an individual basis, perhaps with the assistance of an experienced specialist.

For the combined deep well and open pumping system illustrated in Figure 7.1, the elements of cost will include some or perhaps all of the following elements:

1. Mobilization (lump sum, per unit time on rentals) (a) Permits and submittal preparation (b) Rental or purchase of dewatering equipment (i) Wellscreen and casing (ii) Well pumps, controls, wiring, discharge column (iii) Sump pumps, controls, wiring, hose (iv) Discharge manifold (v) Standby generators (vi) Electric distribution cable and switchgear (vii) Flow meters (c) Transportation to and from jobsite (i) Dewatering equipment (ii) Drill rig

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- (iii) Hydraulic crane (iv) Loader/backhoe (d) Electric power drop 2. Installation and Removal (lump sum) (a) Well drilling and development (b) Well filter gravel (c) Drilling fluid additive (d) Observation wells (e) Gravel and geotextiles

for sumps and ditches (f) Cold weather protection (g) Installation labor (with hydraulic crane, truck, loader, and small backhoe as required) (i) Well pumps (ii) Discharge lines (iii) Construction of sumps and ditches (iv) Sump pumps (v) Electric distribution system (vi) Electric pump connections (vii) Electric substation (viii) Standby generators Removal labor (with hydraulic crane as required) (i) | Pumps, discharge lines (ii) Electric equipment (iii) Standby generators (iv) Well abandonment (v) Grouting of sumps and ditches 3. Operation and Maintenance (per unit time) (a) Electric power (b) Generator fuel (c) Maintenance labor (with crane and backhoe as required) (i) | Maintenance of sumps and ditches (ii) | Removal and replacement of pumps (iii) Pump repair (iv) Periodic readings of observation wells, and preparation of reports (d) Maintenance material (1) Replacement pumps and motors. (ti) Repair parts for pumps, controls, generators (iii) Lube supplies for generators (e) Specialist superintendent

(h)

On certain projects extraordinary costs are incurred that can be significant. They have included:

* Treatment of dewatering effluent

e Periodic chemical analysis of effluent

© Special reports to the engineer or to regulating authorities e Artificial recharge

7.9 UNDESIRABLE SIDE-EFFECTS OF DEWATERING

Under certain conditions, the process of dewatering can have side-effects that are harmful to the project under construction, to other facilities nearby, or to the environment. It is important to identify such conditions during the planning stage, so that procedures can be instituted to control any damage. If a problem emerges unexpectedly after construction has begun, the costs of dealing with it escalate, and the project may be delayed. Powers (1985) presents procedures for identifying and evaluating problems of side-effects, and methods for dealing with them.

Improper dewatering such as open pumping under the wrong conditions, can cause damage to the structure being built or to adjacent structures. The principles in Section 7.3 can prevent such avoidable occurrences.

Ground settlement can occasionally be a problem. Lowering the water table increases the effective stress in the soil. The stress

increase is usually modest, and most soils are not affected significantly. But if there are compressible soils in the vicinity, such as peat, organic silt, or soft clays, settlement may occur. Whether the settlement causes significant damage depends on the thickness and consolidation characteristics of the compressible deposit, the depth of drawdown and the duration of pumping, the foundations of the structures within the zone affected, and the type of their construction.

Untreated timber piles or other wooden structures below the water table can be damaged if the dewatering process exposes them to oxygen.

Groundwater supplies in the vicinity may be affected, by temporary reduction in the yield of supply wells, by salt water intrusion, or by the expansion of contaminant plumes.

The delicate ecological balance of wetlands can be upset by dewatering, particularly if pumping continues for an extended period. Trees or other plantings in urban parks may be affected.

Dewatering discharge sometimes contains substances that make treatment necessary before the water can be released into the surface environment. Natural substances include hydrogen sulfide and methane. Commonly encountered man-made contaminants include volatile organics, petroleum products, acid wastes, and sanitary sewage.

When problems from side-effects such as the above have been identified, their potential severity should be evaluated and procedures planned to deal with them. It may be necessary to control the water without dewatering, using one of the cutoff or exclusion methods (Section 7.3). Sometimes it is cost-effective to accept minor building damage from settlement and arrange for its repair. Temporary replacement water supplies can be provided to groundwater users. Artificial recharge can be employed to restrict the influence zone from the pumping (Powers, 1981, 1985).

7.10 PERMANENT DEWATERING SYSTEMS

Permanent pumping systems are employed for many purposes, including relieved structures such as drydocks, building foundations, and sewage treatment tanks; to prevent seepage into leaking structures; and to contain or recover pollutants in groundwater. The design of permanent systems is similar to that for temporary construction dewatering systems but there are significant differences. Comparison of cost of construction versus operation and maintenance is necessary to determine the optimum balance. Typically, better-quality construction can be justified by long-term savings.

Materials of Construction are selected for long life and low maintenance. A clear understanding of the groundwater chemistry, including potential variations with time is essential. Samples should be recovered during the pumping test, and analyzed at a laboratory experienced in groundwater problems. Piping and wellscreens are frequently made from PVC or ABS plastics, but if the system is to recover solvents from the ground such materials are unsuitable. Fittings, couplings, and valves should be selected on the basis of the chemistry as well as their function.

Pumps are designed for efficient operation, to reduce power costs, and also to avoid surging and cavitation that might cause maintenance and repair problems. Pumping test data must be extrapolated to conditions that can be expected during extended pumping. The impeller, volute, and other parts in contact

with the water must be of suitable materials. Pumps are available in cast iron, plastic, fiber glass, bronze of various alloys suitable to different environments, and stainless steel. If solvents are present in the water, the elastomers in the mechanical seal

must be of viton or some material resistant to the particular substance.

Automatic Controls for permanent systems can be complicated if the purpose is to be accomplished effectively. Many systems operate intermittently to control the water table within design limits. Water-level probes installed in an observation well initiate the starting and stopping of the pump. With systems of several wells, there may be multiple probes that start one pump, then start another if the water level continues to rise. Often a high-level alarm probe is used with a remote signal to alert maintenance personnel to a developing problem.

Encrustation can be a problem with permanent systems. It is a function of the groundwater chemistry and bacteriology, and should be investigated during the pumping test. Agents that have created problems include high hardness, which can result in precipitation of calcium carbonate; iron and manganese, particularly in the presence of *Crenothrix* or other iron-fixing bacteria; and organic slimes, sometimes encountered near sewage treatment plants.

Where there is encrustation potential, special features are advisable. Wells or wellpoints are designed for low entrance velocity, to retard the encrustation. Wellheads are arranged for easy access for acid treatment, cleaning, and redevelopment. Provision is made for safe handling and disposal of the acid. Experience shows that encrustation can be more readily removed if the work is done on a timely, periodic basis. If the condition advances to the stage where the encrustation is developing in the soil outside the wellscreen, acidization may be ineffective and the well has to be replaced. Similarly, if encrustation buildup in a pump advances too far it can cause mechanical damage.

The tops of the wells or wellpoints and the connecting piping are typically buried, to protect against weather and vandalism or for aesthetic reasons. Provision is necessary for access for well inspection, adjustment, and maintenance. Manholes, curb boxes, or pitless adaptors have been employed.

7.11 GROUND FREEZING

In the ground-freezing process (Sanger, 1968; Maishman, 1975) heat is pumped from the ground through freeze pipes until the pore water freezes, forming a cutoff and ground support around an excavation. A great variety of frozen structures can be formed (Maishman and Powers, 1982). Since freezing performs a dual function, it is cost-effective on many projects, particularly small, deep excavations and tunnels.

The shaft in Figure 7.11 is a typical application. Refrigerated brine is circulated through the freeze pipes (Fig. 7.12), which are typically arranged in series/parallel loops. As heat is removed, isotherms move outward in a manner very similar to drawdown contours around a water well. Heat pumping is

analogous to pumping groundwater, and the mathematical relationships used for the analysis are essentially the same. When the 32°F isotherms intersect, the frozen diaphragm is complete. As the process continues the frozen structure increases in thickness. An observation well in the center will exhibit measurable rises in water level as the freeze moves inward. Temperature monitor holes enable observation with thermocouples to confirm that the wall has reached its design strength before excavation begins. The freezing is continued until the excavation is complete and the shaft has been lined. If the inner surface of the frozen wall is to be exposed to ambient conditions for an extended period, an insulating blanket is installed.

The brine is chilled by portable refrigeration plants, typically of 100 to 300 horsepower. On larger projects a group of such

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Freeze pipe

Isotherms

PLAN

Brine return

Noe Brine supply of

Freeze pipe

Frozen earth

Subgrade

a

SECTION Fig. 7.11 Ground freezing.

plants may be employed, with a central brine-circulating system. The heat is released to the atmosphere through an evaporative condenser, which requires a fresh water supply of about 5 to 10 gpm.

Quality Control is essential to accomplishing the desired result on schedule. Freeze pipes are spaced closely, typically 4 to 6 ft on centers, in order to achieve the design thickness within a reasonable formation period. If a borehole drifts out of plumb during drilling, the design spacing may not be achieved at depth. It is good practice to survey each hole with a directional inclinometer. If spacing between two pipes exceeds the allowable,

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Brine return

Brine supply

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Freeze pipe

Fig. 7.12 Brine circulation.

an additional pipe is installed. Each freeze pipe must be hydrostatically tested, since any brine leakage would impair the freeze. Appropriate instrumentation includes an array of temperature-monitoring holes and temperature-monitoring points in the brine system, as well as the instruments on the refrigeration plant.

Groundwater Movement in the vicinity of the freeze can create problems if the velocity exceeds 3 or 4 ft per day. The movement presents an additional heat load on the freezing system. Natural velocities over 3 ft per day are unusual, but transients can occur, for example, in the flood plain of a volatile river. When ground freezing is being considered, it is advisable to investigate whether any groundwater is being pumped nearby.

Liquid Nitrogen is another process that has been employed for ground freezing. The cost per unit of heat extracted is much higher than with refrigerated brine, but nitrogen can be cost-effective on small projects of short duration that are located near a manufacturing plant producing the fluid. The nitrogen is stored in a vacuum-insulated vessel on the site and is expended in the process. Special trucks refill the vessel periodically. The gas is nontoxic, but precautions are necessary to prevent its accumulation in enclosed areas. On a liquid-nitrogen installation, freeze pipes are typically smaller in diameter than with brine, and more closely spaced. Metallic hoses and other

special devices and materials are used to withstand the extremely low temperatures, on the order of -320°F .

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